Jenkins

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What is CI / CD ?

- CI/CD stands for **Continuous Integration / Continuous Deployment**
- Goal of CI/CD is to "release early and often". And the way to do this, is by **automating the whole software release cycle**

Continuous Integration

- Means new code changes are continuously built, tested and merged to shared repository

Continuous Deployment

- Automating further stages of the pipeline
- Automatically deploying to different deployment environments including releasing to production



- Build automation is a basic component of CI/CD pipelines

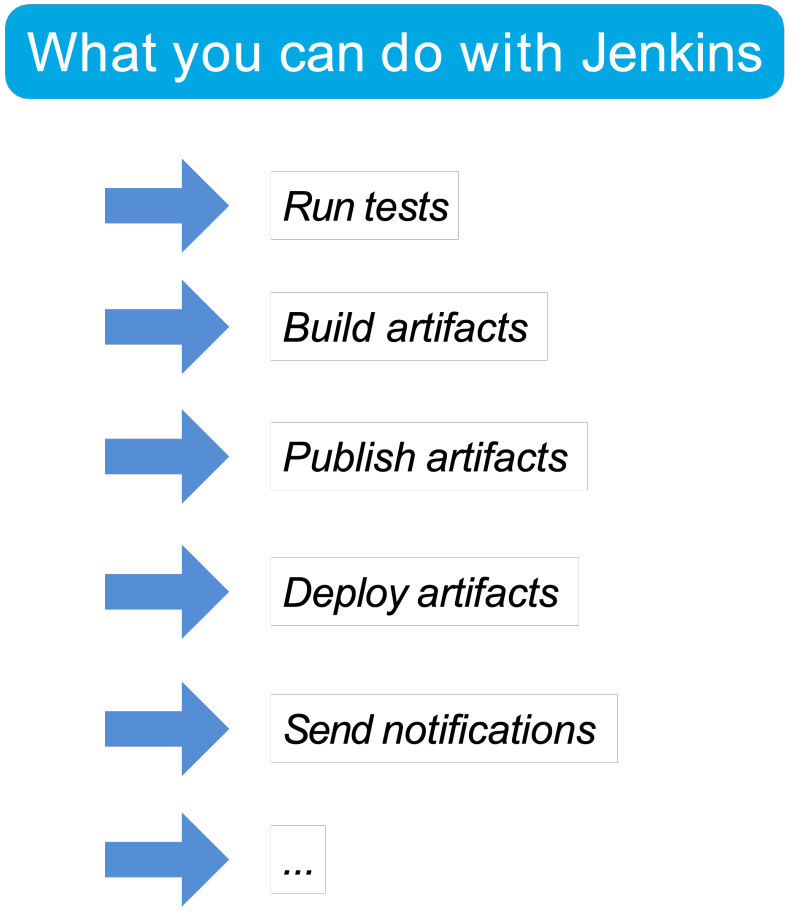
Jenkins = most used in the industry

- Jenkins is currently one of the most used build automation CI/CD tool in the industry



Introduction to Jenkins - 1

1. Jenkins is a software that you **install on a dedicated server**
2. It has a **UI** to configure your builds
3. You need to **install all the tools** you need to get the tasks done (Docker, Gradle/Maven/npm, etc.)
4. Then you need to **configure the tasks** (run tests, build app, deployment etc.)
5. And **configure the automatic trigger** of the workflow

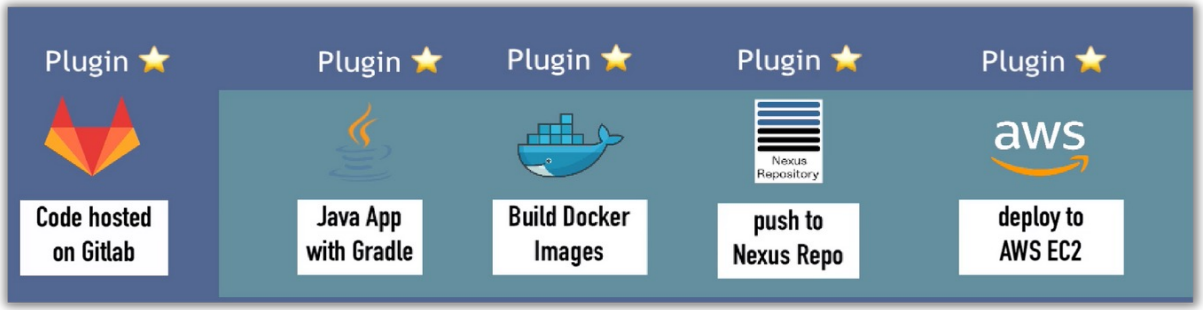


Introduction to Jenkins - 2

One of most important capabilities a build automation needs to have is integration with other technologies

Jenkins makes this integrations possible with **Plugins.** Plugin = **extension to Jenkins functionality**

There are plugins for Docker, Build Tools, Repositories, Deployment Servers etc.



Introduction to Jenkins - 3

- **DevOps Engineer task is to administer Jenkins** (plugins, setup etc.), create or help developers to set up build automation jobs in Jenkins

How it works:

RUN TESTS — Build Tools need to be available

BUILD ARTIFACTS — Build Tools or Docker need to be available

PUBLISH ARTIFACTS — Store credentials in Jenkins



Jenkins User must have access to all these technologies and platforms

What to configure:

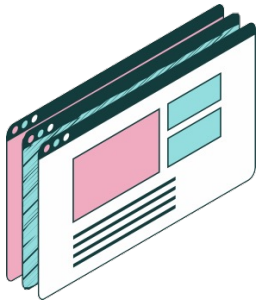
- Install Jenkins and prepare everything
- The setup needs to be done only once
- Plugins, Credentials etc. can be used for different projects

Install Build Tools in Jenkins

Jenkins is used to **build applications**

So first we need to **install and configure the build tools** in Jenkins

2 ways to install and configure those tools:



(1) Jenkins Plugins

Just install plugin (via UI) for your tool



(2) Install Tools directly on server

Enter in remote server and install Inside the Docker container, when Jenkins runs as container

Jenkins Job Types

Freestyle job

- **To orchestrate simple jobs for a project**
- **Configuration: Via UI**
- **Very limited, limited to the input fields of plugin**

Pipeline job

- **For more complex workflows, so suitable for CI/CD pipelines**
- **Ability to divide the tasks into different stages**
- **Configuration: UI and Scripting (Pipeline as Code) Much more powerful**

More advantages or use cases for Pipeline job:

You want to execute 2 tasks in parallel

Usage of user input or conditional statements Set variables

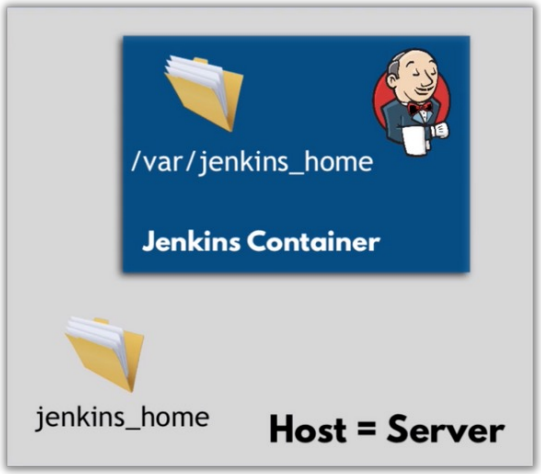
Jenkinsfile

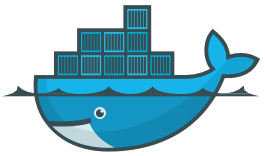
```
CODE_CHANGES = getGitChanges()
pipeline {
    agent any
    stages {
        stage("build") {
            when {
                expression {
                    BRANCH_NAME == 'dev' && CODE_CHANGES == true
                }
            }
            steps {
                echo 'building the application...'
```

Jenkins Installation

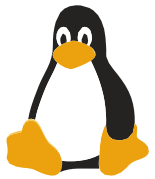
2 ways to install Jenkins on a server:

1) Run Jenkins as Docker container





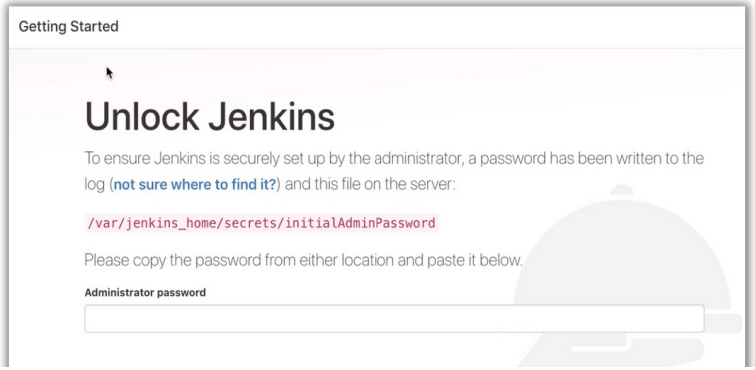
2) Install Jenkins directly on OS

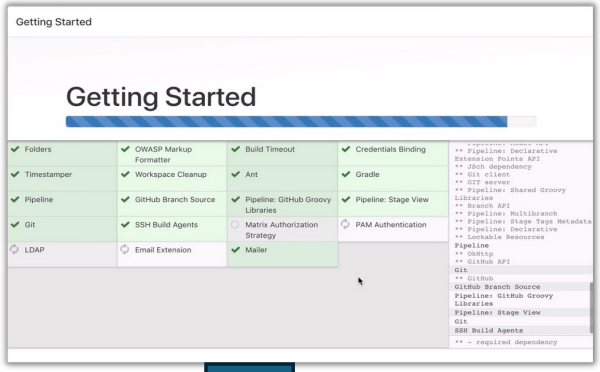


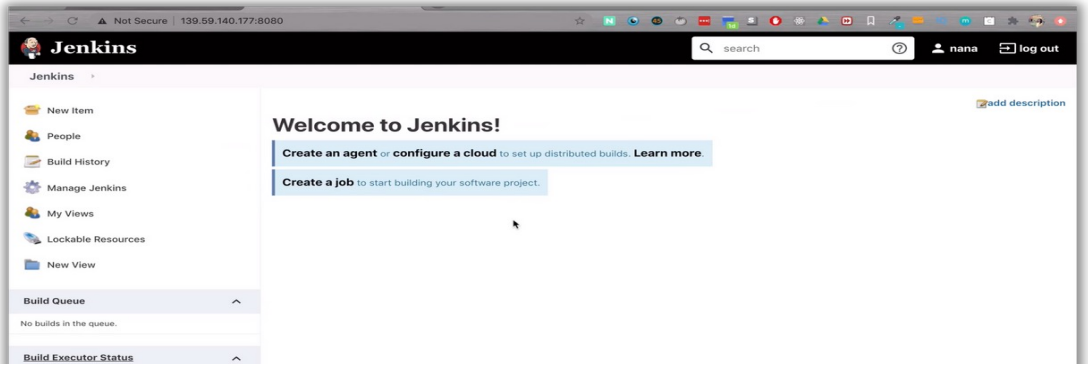
```
Bootcamp — root@ubuntu-s-2vcpu-2gb-fra1-01: ~ — ssh root@207.154.235.187 — 95×26
root@ubuntu-s-2vcpu-2gb-fra1-01:~# docker run -p 8080:8080 -p 50000:50000 -d \
> -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
```

Getting Started

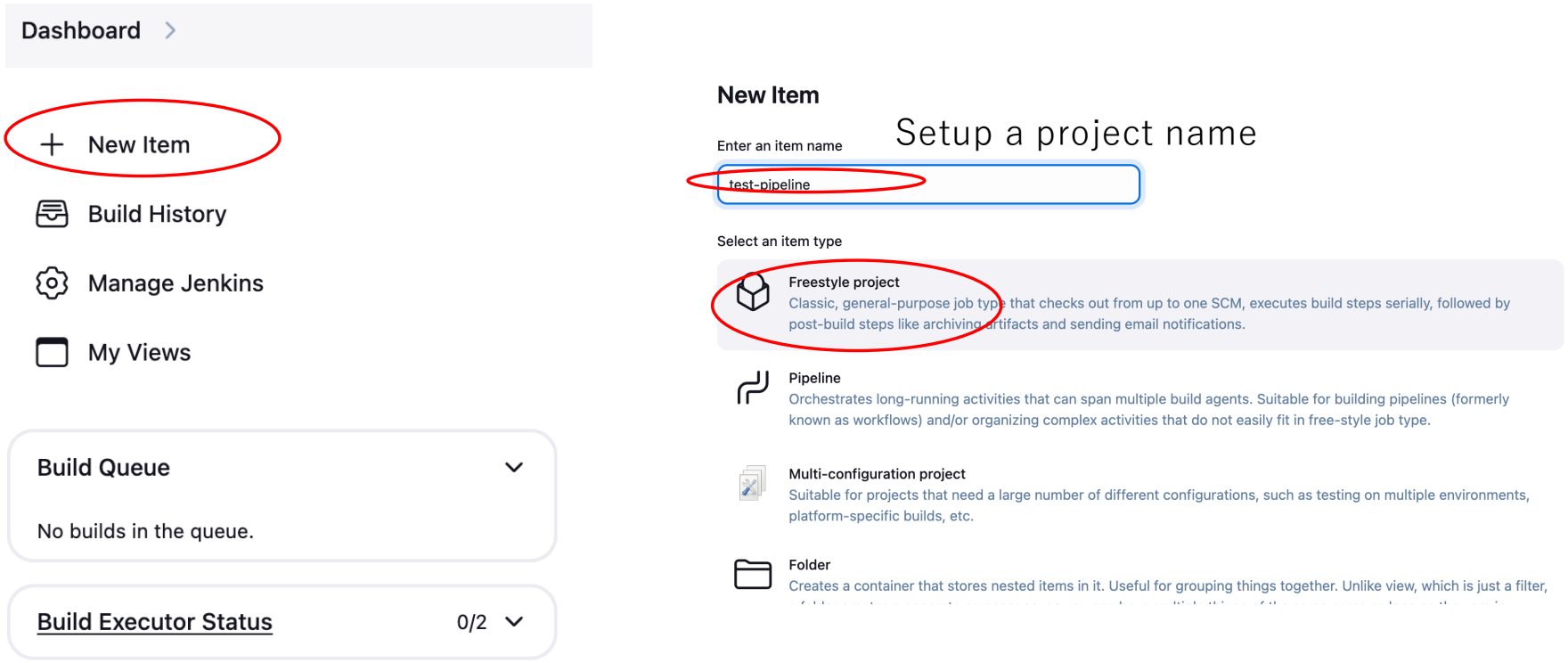
- Install the Jenkins to EC2
  - https://www.jenkins.io/doc/book/installing/linux/#debianubuntu





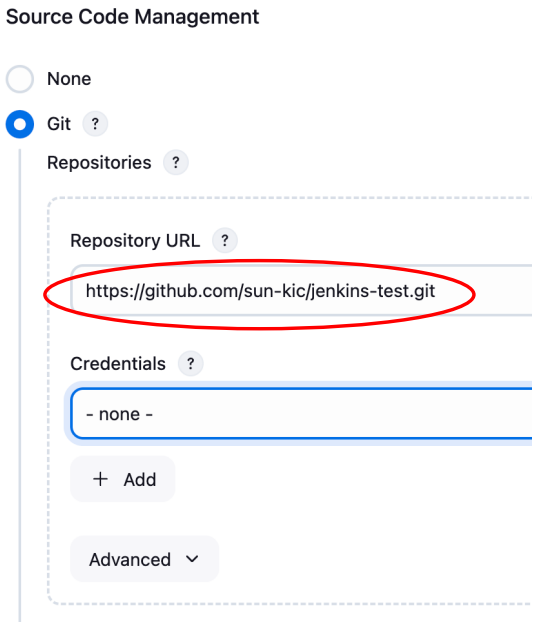


Start a new project



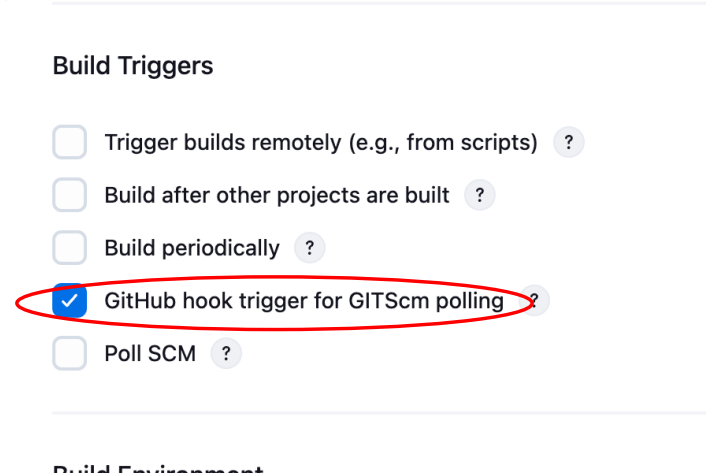
Link with the github

- Create a new github repository and copy the link to the Jenkins project

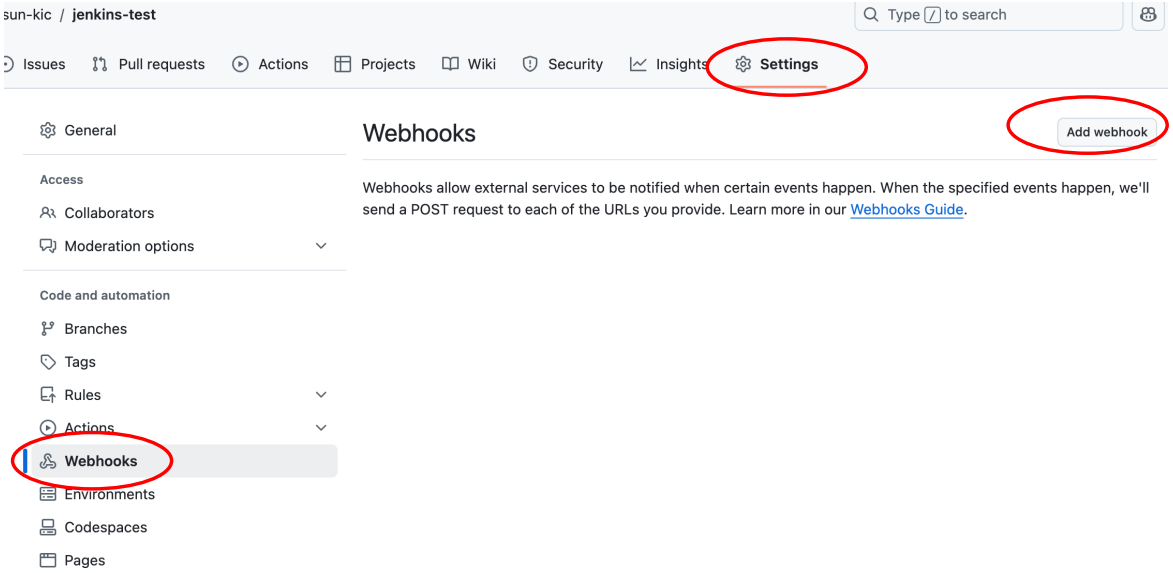


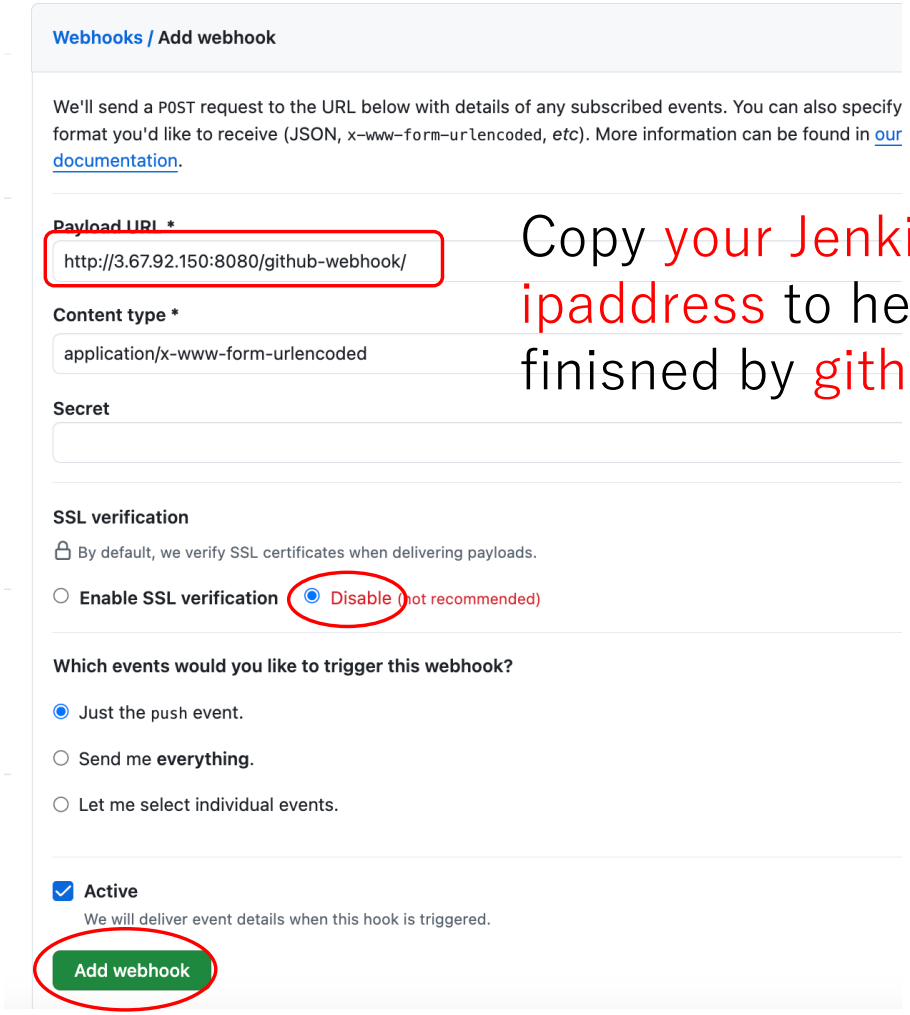
Use the webhooks to build github trigger

Check the build triggers in [GitHub hook trigger for GITScm polling]



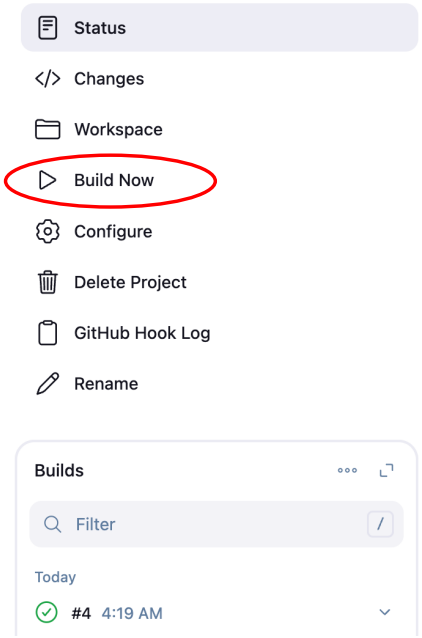
Add a new webhook from github repository

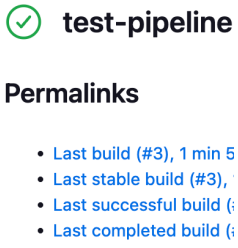




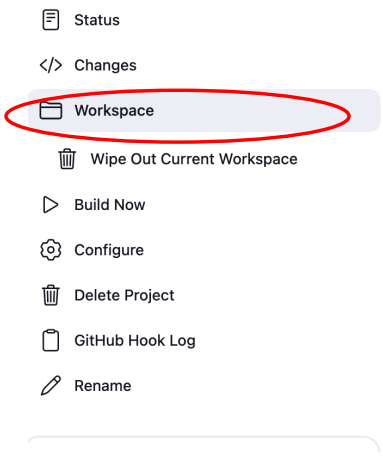
Copy your Jenkins server ipaddress to here and finisned by github-webhook/

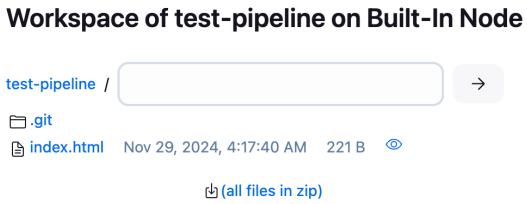
Check the first build







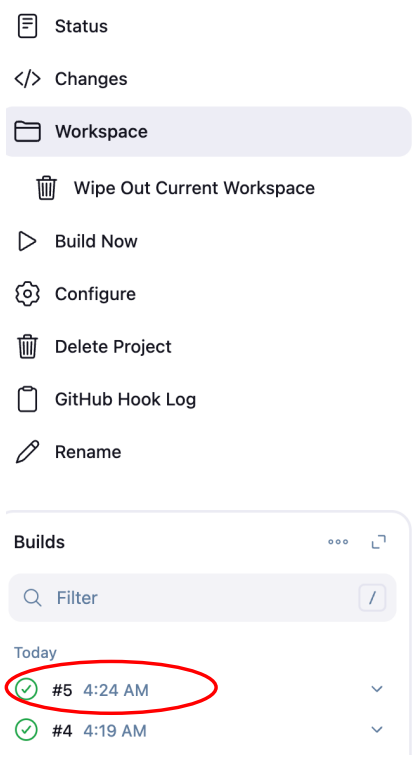




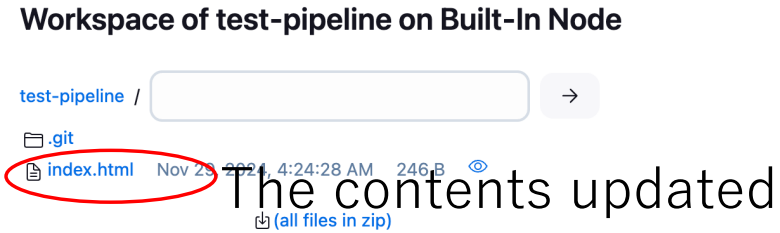
You can find the file from workspace

Check the push event is work

- Edit the index.html and push to github again.

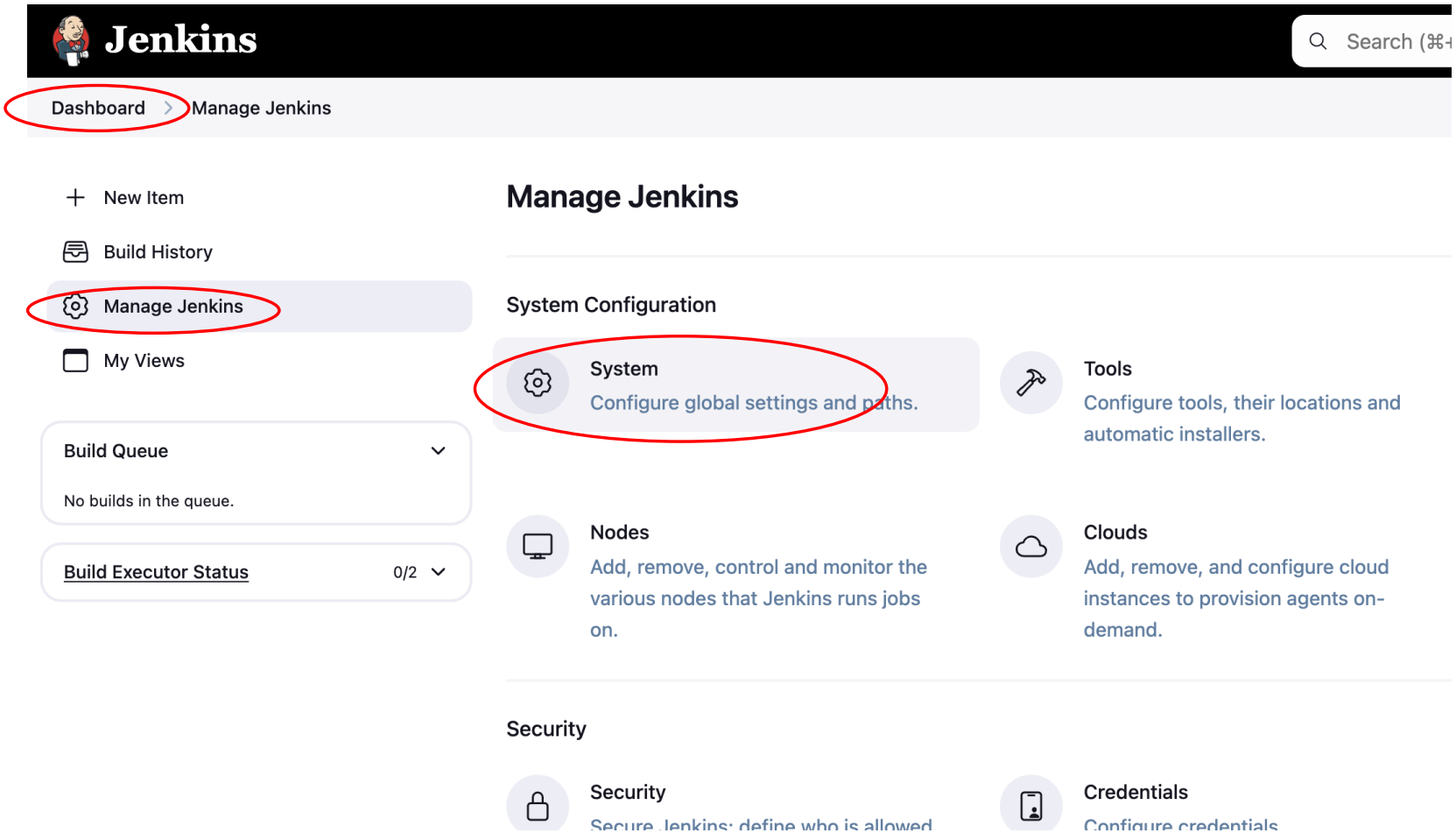


A new build happened



The contents updated

Setup the docker server in Jenkins



Use Docker to build the website

Step1: Install the docker

Step2: Add the jenkins user to docker group

```
sudo usermod -aG docker jenkins
sudo systemctl restart jenkins
```

Step3: Add a Execute shell in pipeline configuration

